Capstone Development Reflection Journal

Module B: AI for Leaders

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Weeks 2–6 · artifacts included

Week 2

Assignment/Capstone work: I picked three industries: finance, environment, and entertainment. I am a valuation specialist at a hedge fund in Hong Kong, and valuing derivatives is my job and the subject I care about most. Environment is in because the EV transition is the industrial change that has mattered most in my lifetime. Entertainment is music and film, which I know as a consumer.

Artifact (required): three industries, two public datasets each.

Industry	Dataset A	Dataset B
Finance	Historic options (Dubach): SPY/IWM/QQQ chains, 2008–2025	FINRA equity short interest (biweekly files)
Environment	CPCA China passenger-vehicle sales, Jan 2018–Apr 2024 (Kaggle / felixzhao)	National CO2 panel (later dropped; too far from a decision I would own)
Entertainment	Ultimate Spotify Tracks DB (Hamidani, Kaggle)	The Movies Dataset (Banik, 2017, Kaggle / TMDB)

Reflection: Narrowing to three industries was the first decision I had to make. I wanted the most technically impressive datasets. Choosing industries first changed the question: which problems would I still care about after the grade is posted? Markets, the energy transition, and culture.

Week 3

Assignment/Capstone work: I extracted the first data: the SPY options snapshot for 31 December 2024 (9,646 contracts, 23 columns) and the CPCA China auto file (38,806 model-

month rows). Problems spotted immediately: 625 zero-bid option contracts (dead quotes), duplicate model-month rows, and “NA” strings in the body-type column.

Artifact (required): sample rows from each of the five datasets that ended up in Milestone One, with a one-line description. Full column lists are in the paper.

Dataset 1 — SPY options, 31 Dec 2024. One listed contract. Near-the-money sample (the file’s first rows are deep in-the-money expiries).

contract_id	strike	type	last	IV
SPY250207C00594000	594	call	7.80	0.132
SPY250207P00594000	594	put	12.94	0.142
SPY250207C00595000	595	call	7.39	0.132

Dataset 2 — CPCA China passenger-vehicle sales. One model in one month.

year_month	brand	model	units	is_ev
2022-11	Tesla	Model Y	69,098	EV
2022-11	Wuling	Hongguang MINIEV	68,567	EV
2022-11	BYD	Song PLUS	64,145	EV

Dataset 3 — Spotify tracks. One track, audio features plus genre.

artists	track	popularity	danceability	genre
Gen Hoshino	Comedy	73	0.676	acoustic
Ben Woodward	Ghost - Acoustic	55	0.42	acoustic
Ingrid Michaelson;ZAYN	To Begin Again	57	0.438	acoustic

Dataset 4 — TMDB movies. One film. Grumpier Old Men shows a zero budget that is missing data, not a free film.

title	release	budget	revenue	vote
Toy Story	1995-10-30	30,000,000	373,554,033	7.7
Jumanji	1995-12-15	65,000,000	262,797,249	6.9
Grumpier Old Men	1995-12-22	0	0	6.5

Dataset 5 — FINRA short interest, 30 Sep 2026. One stock on one settlement date. File: shrt20260930.csv.

settlementDate	symbolCode	short shares	changePercent	daysToCover
2026-09-30	A	5,240,309	3.18	1.78
2026-09-30	AA	11,725,447	-2.16	2.85
2026-09-30	AAA	7,957	456.82	1.00

Reflection: Data quality is the analysis. The zero-bid contracts looked ordinary until I asked what a zero bid means: nobody will buy at any price. Deleting 6.5% of the data before modeling felt drastic. Training on prices that do not exist is worse.

Week 4

Assignment/Capstone work: I wrote down the professional problems that actually attach to these files, then added a fifth dataset that sits next to the options work instead of repeating it. The professional problems. I work as a valuation specialist at a hedge fund in Hong Kong. The files I kept are the ones that map onto questions I already have to explain, or that I care about as a consumer. In finance the live problem is not downloading prices. It is whether a quoted smile and a set of Greeks are internally consistent, and whether a short-interest file is describing crowded positioning or names that simply do not trade. A wrong IV surface mis-marks a book. A short-interest spike that is actually a split creates a false squeeze alarm for risk and for stock loan. In the auto market the problem is planning: which segments electrified, and whether a subsidy cliff changed that. Inventory and battery supply are expensive if the mix is wrong. In music and film the problem is allocation. Streamers and studios spend against noisy popularity and revenue numbers. A popularity label that is really a genre-balancing artifact, or a movie budget coded as zero, will send a model the wrong way. Milestone Two should stay attached to those questions, not to a generic accuracy score.

Artifact (required): a new related dataset, with a sample and a public link. I added FINRA's 30 September 2026 equity short-interest file (shrt20260930.csv). It is one date and 22,581 names. It

sits next to the SPY chain: listed-market microstructure on one side, positioning and cover time on the other. No price join.

Link: https://cdn.finra.org/equity/otcmarket/biweekly/shrt20260930.csv

settlementDate	symbolCode	issueName	currentShortPositionQuantity	daysToCoverQuantity
2026-09-30	A	Agilent Technologies Inc.	5,240,309	1.78
2026-09-30	AA	Alcoa Corporation	11,725,447	2.85
2026-09-30	AAA	Alternative Access First Prior	7,957	1.00

Reflection: The module treats predictive modeling as supervised segmentation, which changed how I look at market data. An option chain is a segmentation problem: the model learns boundaries across moneyness and expiry the way a tree segments customers. After that, the industry label mattered less to me than the shape of the problem.

Week 5

Assignment/Capstone work: I locked the five datasets: SPY options (finance), China EV sales (environment), Spotify tracks and TMDB movies (entertainment), and the FINRA 30 September 2026 short-interest file (finance). Kaggle would not open from my machine, so Spotify and TMDB came from local copies of the files Kaggle publishes. I built a slideshow that presents each dataset with a sample, why it has value, why I kept it, the questions it can answer, and a decision-tree / linear-discriminant demo on Dataset 5. I tried a SpaceX launches file for environment. It was too small, so I left it out.

Artifact (required): slideshow of the five datasets. The nine slides are printed below so this PDF does not depend on a separate download. The same file lives in the course repo as milestone1/dx699-week5-datasets.pptx. This is the same one-date FINRA file used in the

paper (shrt20260930.csv: 22,581 names by 14 columns on 30 September 2026), not a stacked 90-date panel.

Slide 1 of 9



Slide 2 of 9

The five datasets

#	Dataset	Industry	Size	Unit
1	SPY options, 31 Dec 2024	Finance	9,646 x 23	One listed contract
2	China passenger-vehicle sales (CPCA)	Environment	38,806 x 10	One model-month
3	Ultimate Spotify Tracks DB	Entertainment	114,000 x 20	One track
4	The Movies Dataset (TMDB)	Entertainment	45,466 x 24	One film
5	FINRA short interest, 30 Sep 2026	Finance	22,581 x 14	One stock, one date

Three industries. Mixed numeric, categorical, text, and nested JSON. Dataset 5 is one FINRA settlement date, not a stacked panel.

Slide 3 of 9

1 SPY equity options

Value: a full listed chain with quotes, volume, open interest, IV, and Greeks in one table.
Why it is interesting to me: I value derivatives at a Hong Kong hedge fund. This is a direct empirical check of the volatility smile.
Questions: Can a model recover IV from moneyness and expiry? Do the Greeks behave as the textbooks say? What drives IV more, moneyness, expiry, or liquidity?

Sample (first three rows used in the paper)

contract_id	strike	type	last	IV
SPY250207C00594000	594	call	7.80	0.132
SPY250207P00594000	594	put	12.94	0.142
SPY250207C00595000	595	call	7.39	0.132

Slide 4 of 9

2 China passenger-vehicle sales

Value: official CPCA registrations through the EV transition, with brand, body, price band, and an EV flag.
Why it is interesting to me: the demand side of the largest industrial shift I have lived through.
Questions: Which attributes predicted electrification? Did the January 2023 subsidy end change price elasticity? Where does a seasonal baseline break?

Sample (first three rows used in the paper)

year_month	brand	model	units	is_ev
2022-11	Tesla	Model Y	69,098	EV
2022-11	Wuling	Hongguang MINIEV	68,567	EV
2022-11	BYD	Song PLUS	64,145	EV

3 Spotify track features

Value: 114,000 tracks with audio features and a popularity score, balanced across 114 genres.
Why it is interesting to me: I listen to this market every day and want to know which sonic attributes it rewards.
Questions: Which features predict popularity? Do tracks cluster into stable archetypes for playlists?

Sample (first three rows used in the paper)

artists	track	popularity	danceability	genre
Gen Hoshino	Comedy	73	0.676	acoustic
Ben Woodward	Ghost - Acoustic	55	0.42	acoustic
Ingrid Michaelson;ZAYN	To Begin Again	57	0.438	acoustic

4 TMDB movie metadata

Value: budget, revenue, votes, free text, and nested JSON for cast, crew, and genres.
Why it is interesting to me: as a film consumer I want to know what the market actually pays for before a studio spends.
Questions: What drives revenue? Can a pre-release hit/flop classifier flag failures?

Sample (first three rows used in the paper)

title	release	budget	revenue	vote
Toy Story	1995-10-30	30,000,000	373,554,033	7.7
Jumanji	1995-12-15	65,000,000	262,797,249	6.9
Grumpier Old Men	1995-12-22	0	0	6.5

5 FINRA equity short interest

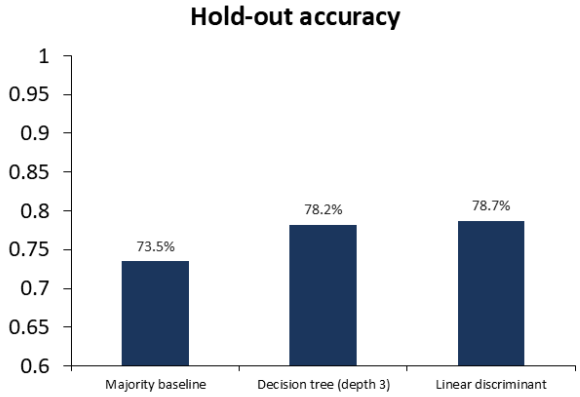
Value: 22,581 names on 30 Sep 2026 with short shares, average daily volume, and days-to-cover. One file, one date.
Why it is interesting to me: squeeze risk and borrow demand are live issues in a hedge-fund seat.
Questions: Which names are heavily shorted relative to volume? Does counting shares vs days-to-cover change who looks crowded?

Sample (first three rows used in the paper)

settlementDate	symbolCode	short shares	changePercent	daysToCover
2026-09-30	A	5,240,309	3.18	1.78
2026-09-30	AA	11,725,447	-2.16	2.85
2026-09-30	AAA	7,957	456.82	1.00

Demo: decision tree vs linear discriminant

Task: flag a name as crowded when days-to-cover > 5 on the 30 Sep 2026 FINRA file. Features: log average daily volume and market class. Zero-volume names dropped (n = 19,438). Held-out 30%. Majority class is 73.5% not-crowded.



What the tree uses
First split: log(ADV) ≤ 6.20, i.e. volume under about 500 shares a day. Low-volume OTC names are the crowded class.

Tree accuracy 78.2%. LDA 78.7%. Both beat the 73.5% majority baseline by about five points.

Week 5 point: a linear discriminant and a shallow tree find the same volume story. Neither is a trading signal. Both show the label is partly a liquidity label.

Slide 9 of 9

Used in Milestone One Sections 1 to 4

Each dataset has a description, a data-quality problem, one or two learning types, and industry questions.
Dataset 5 is the FINRA file shrt20260930.csv, not a 90-date stack.

Open the .pptx on GitHub.Demo in the deck: crowded = days-to-cover above 5 on the FINRA file, predicted from log volume and market class (n = 19,438 after dropping zero volume).

Majority baseline 73.5%. Depth-3 tree 78.2%. Linear discriminant 78.7%. The tree's first split is log(ADV) ≤ 6.20 (about 500 shares a day). Low-volume OTC names are the crowded class. That is a liquidity story, not a trading signal.

Reflection: I spent more time dropping datasets than adding them. Letting go of the CO2 panel hurt. The data was beautiful. The rubric rewards fit over volume: five datasets I can defend beat eight I cannot. Week 5's linear discriminants also made the FINRA demo useful: a shallow tree and an LDA both recover the same volume split, and both only beat the majority class by about five points.

Week 6

Assignment/Capstone work: I audited data quality on the three later files and checked, for each dataset, whether regression and classification are actually feasible. Spotify had 450 duplicate rows and 24,259 duplicate track ids (label noise). TMDB had 17 duplicates and zero-coded budgets. The FINRA file (one date) has a blank ticker (Nano Labs), days-to-cover floored at 1.00 on 8,295 rows and capped at 999.99 on 4,034 names, and 3,143 rows with zero average daily volume. I also restructured the paper by section, with the rubric's exact labels, double spacing, and one page of examples.

Artifact (required): per-dataset regression / classification feasibility.

Dataset	Regression feasible?	Classification feasible?	Note
1 SPY options	Yes: quoted IV from moneyness and expiry	Yes: liquid vs illiquid from spread, volume, OI	625 zero-bid quotes must be dropped first
2 China EV sales	Weak: monthly EV units from lags is a time series; rubric discourages it	Yes: EV vs gasoline from brand, body, price	Prefer the classifier
3 Spotify	No natural numeric target I would stand behind	Yes: popularity tier or genre after de-dupe	24,259 repeated track ids are label noise
4 TMDB	Yes: log revenue from budget, genre, timing	Yes: hit vs flop from pre-release features	Zero budgets are missing, not zero
5 FINRA 30 Sep 2026	Yes: log short shares from log volume and market class (R^2 0.42)	Yes: days-to-cover > 5 from volume tier and class	Do not put short shares or days-to-cover in the classifier features

Reflection: The Spotify label noise was the best puzzle this week. Identical features with contradictory genre labels break naive classification, and the fix changes the problem statement. The other lesson was formatting: double spacing halves the words on a page.